

Have Some Fun Converting Plain Text to Handwritten Text

Recurrent neural networks (RNN) can be used to generate handwritten text. This is a fun project that you might like to try out for yourself. This article details the steps to be taken to generate handwritten text from plain text.



Typing has brought a lot of ease to content generation, but sometimes, you may need handwritten text rather than the typed version. Cursive fonts try to address this but fail since the nature of the problem is not addressed; instead, only the symptoms are treated.

Alex Graves has published a paper which demonstrates how to generate sequences with recurrent neural networks (RNN). It can be found at <http://arxiv.org/abs/1308.0850>. The long and short of it is that we can now generate handwriting using RNNs. In this article, we demonstrate a nifty little program which converts any plain text file into a series of images that have handwriting on them instead of the usual printed text.

This article does not aim to discuss Graves' paper, but rather, lets you use the research and create your own handwritten text. So, let's dive in.

This has been tested on Arch Linux and Ubuntu 15. It should work on any platform where Python3.x is installed. The steps are as follows.

First off, we need to install what's required to run this software. Let's first install Python3 from <http://python.org>.

If you are on a Linux system, Python is already installed. On Ubuntu, however, you will need the correct version of Python and so need to run the following command:

```
sudo apt-get install python3
```

Once that is done, we need to install some of the packages required for our software. To avoid complicating matters, let's use the virtual environment system for Python. To create a virtual environment, we run the following command:

```
virtualenv -p python3 env
```

After this, to activate the virtual environment, use the command given below:

```
source env/bin/activate
```

Now that the virtual environment is created and active, we can install libraries in it using the following command: